Institute of Technology of Cambodia
ECAM LaSalle

Name: Sovan Chhorponn
ID: e20220873

Network and Security
Assessment I

I. Questions

1. What is the difference between IP address and Mac address?
2. Among all network topologies, which topology that causes much more expense when we use the same amount of the nodes? Why?
3. Among 7 layers of OSI model, which layer that provide the security for network communication? Describe its function

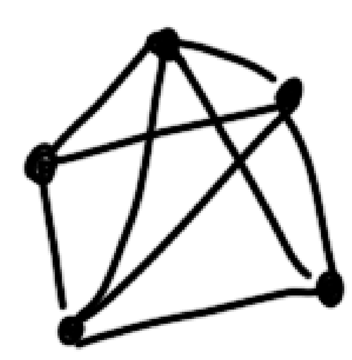
II. Exercises

4. There are a new startup company and it consists of 5 departments. Each department is needed to design with required host respectively: 200, 15, 32, 64 and 10 hosts.
   ➔ Make IP calculation to assign in this new company

Solution

1. The difference between IP address and Mac address is:
- The IP address helps in identifying the connection of a network. It is an address that assigned to our device when we are using Wifi or network so it can communicate with other devices on the same wifi or network. If we change wifi or network, the IP address also changes.
- Mac address is the hardware/Physical address of the device. It does not change when we move it to another wifi or network. Every Mac address is difference and was created by Manufacturer.

2. Among all topologies when we use the same amount of nodes, the topologies that cause the most expense is the mesh topology because the mesh topology requires all the nodes to connect to each other unlike any other topology so it needs more maintenance, more computer power, more wires/Cable.



4. The layer that provide the security for network communication is the presentation layer (Layer 6).

It defines the data format and encryption. (Compress & Decompress, Encryption & Decryption)

4. New start-up company, need 5 department and each department needs number of host 200, 15, 32, 64 and 10 hosts.

Let's make IP calculation.

decending order

$$\left.\begin{array}{l} \text{Department 1} : 200 < 2^8 \\ \text{Department 4} : 64 < 2^7 \\ \text{Department 3} : 32 < 2^6 \\ \text{Department 2} : 15 < 2^5 \\ \text{Department 5} : 10 < 2^4 \end{array}\right\} < 2^{16}$$

Class B

I choose $172.69.0.0/16$

$D_1 = 200 < 2^8 - 2$

$I_p = 2^{32 - \text{Subnet mask}} = 2^8$

$\Rightarrow$ Subnet mask = 24

$\Rightarrow$ Subnetwork = $2^{24-16} = 2^8 = 256$

$\Rightarrow$ Department 1

Subnet$_0$ = IP_{net} = $172.69.0.0/24$

IP_{usable} = $172.69.0.1/24 \leftrightarrow 172.69.0.254/24$

$IP_{Broadcast}$ = $172.69.0.255/24$

Subnet$_1$ = IP_{net} = $172.69.1.0/24$

IP_{usable} = $172.69.1.1/24 \leftrightarrow 172.69.1.254/24$

$IP_{Broadcast}$ = $172.69.1.255/24$

Subnet$_2$ = IP_{net} = $172.69.1.0/24$

IP_{usable} = $172.69.1.1/24 \leftrightarrow 172.69.1.254/24$

$IP_{Broadcast}$ = $172.69.1.255/24$

Subnet$_3$ = IP_{net} = $172.69.2.0/24$

IP_{usable} = $172.69.2.1/24 \leftrightarrow 172.69.2.254/24$

$IP_{Broadcast}$ = $172.69.2.255/24$

$\vdots$

Subnet$_{255}$ = IP_{net} = $172.69.255.0/24$

IP_{usable} = $172.69.255.1/24 \leftrightarrow 172.69.255.254/24$

$IP_{Broadcast}$ = $172.69.255.255/24$

Choose Subnet$_1$: $IP_{net} = 172.69.1.0/24$

Department 4 : $64 < 2^7 - 2$

$IP_{network} = 2^{32 - \text{subnetmask}} = 2^7$

$\Rightarrow$ subnetmask = 25

$\Rightarrow$ Subnetwork = $2^{25-24} = 2$

Department 4.

Subnet $_1$: $IP_{net} = 172.69.1.0/25$

$IP_{usable} = 172.69.1.1/25 \longleftrightarrow$

$172.69.1.126/25$

$IP_{Broadcast}: 172.69.1.127/25$

Subnet $_2$: $IP_{net} = 172.69.1.128/25$

$IP_{usable} = 172.69.1.129/25 \longleftrightarrow$

$172.69.1.254/25$

$IP_{Broadcast}: 172.69.1.255/25$

Choose $172.69.1.128/25$

$D_3 = 32 < 2^6 - 2$

IP network : $2^{32 - \text{Subnet mask}} = 2^6$

$\Rightarrow$ subnet mask $= 26.$

Subnetwork $= 2^{26-25} = 2^1 = 2$

Department 3

Subnet $_1$: $IP_{net} = 172.69.1.128/26$

$IP_{usable} = 172.69.1.129/26$

$172.69.1.190/26$

$IP_{Broadcast}: 172.69.1.191/26$

Subnet $_2$: $IP_{net} = 172.69.1.192/26$

$IP_{usable} = 172.69.1.193/26 \longleftrightarrow$

$172.69.1.254/26$

$IP_{Broadcast}: 172.69.1.255/26$

Choose $172.69.1.192/26$

$D_2 = 15 < 2^5 - 2$

IP network : $2^{32 - \text{Subnet mask}} = 2^5$

$\Rightarrow$ subnet mask $= 27$

Subnetwork $= 2^{27-26} = 2^1 = 2$

Department 2

Subnet $_1$: $IP_{net} = 172.69.1.192/27$

$IP_{usable} = 172.69.1.193/27$

$172.69.1.222/27$

$IP_{Broadcast}: 172.69.1.223/27$

Subnet $_2$: $IP_{net} = 172.69.1.224/27$

$IP_{usable} = 172.69.1.225/27 \longleftrightarrow$

$172.69.1.254/27$

$IP_{Broadcast}: 172.69.1.255/27$

Choose $172.69.1.224/27$

$D_5 = 10 < 2^4$

IP network : $2^{32 - \text{Subnet mask}} = 2^4$

$\Rightarrow$ subnet mask $= 28$

Subnetwork $= 2^{28-26} = 2$

Department 5

Subnet $_1$: $IP_{net} = 172.69.1.224/28$

$IP_{usable} = 172.69.1.225/28$

$172.69.1.238/28$

$IP_{Broadcast}: 172.69.1.239/28$

$Subnet_2 : IP_{net} = 172.69.1.240/28$

$IP_{usable} = 172.69.1.241/28 \longleftrightarrow$

$172.69.1.254/28$

$IP_{Broadcast} : 172.69.1.255/28$